The empirical recurrence for OEIS A295845: recovering a conjecture that is not written down

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Abstract

OEIS A295845 records "Empirical recurrence of order 73", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 146$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 73, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 73 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295845 is "Number of n X 6 0..1 arrays with each 1 adjacent to 1 or 2 king-move neighboring 1's.". It has offset 1 and begins

$$21, 789, 9645, 190943, 3804661, 68406015, 1296880337, 24467737897, \ \ldots$$

The entry states:

> Empirical recurrence of order 73 (see link above).

As of the "Last modified" line on the live entry (Jun 22 2022 00:44:45, revision 6) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4160$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 146$ of the 4160, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 146.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 292$ exact terms of the model returns order 73 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{73} c_i\, a(n-i),$$

c_1	16	c_{20}	80782310970	c_{39}	−13130608851971	c_{58}	941917413054
c_2	4	c_{21}	−374101445013	c_{40}	53667319988667	c_{59}	224324803445
c_3	1412	c_{22}	−1924729563355	c_{41}	113703127439421	c_{60}	−510576399262
c_4	−9949	c_{23}	773314866642	c_{42}	−42721611294221	c_{61}	−198631598555
c_5	28509	c_{24}	2697879192493	c_{43}	−33825392101548	c_{62}	167584357222
c_6	−446021	c_{25}	4231230332264	c_{44}	−12061793236365	c_{63}	86928864815
c_7	1530467	c_{26}	−1558526354862	c_{45}	−99300872175410	c_{64}	21495454848
c_8	−1771173	c_{27}	−14188854282059	c_{46}	85040390275904	c_{65}	4117732703
c_9	24781787	c_{28}	4422593061136	c_{47}	4985976169213	c_{66}	−2974357588
c_{10}	−64015068	c_{29}	13213916133462	c_{48}	22667862743856	c_{67}	−903591010
c_{11}	−11069672	c_{30}	36297650028292	c_{49}	−18270641727464	c_{68}	−80247599
c_{12}	−224743073	c_{31}	−13660881409272	c_{50}	−51442902135890	c_{69}	14250030
c_{13}	206234393	c_{32}	−43728695787539	c_{51}	36200320481366	c_{70}	−1628426
c_{14}	3670362326	c_{33}	−19732983166356	c_{52}	10437221343049	c_{71}	81084
c_{15}	−3162519000	c_{34}	−8743896307888	c_{53}	3941156949047	c_{72}	−21604
c_{16}	5323684825	c_{35}	39219757895482	c_{54}	−6806275418618	c_{73}	−342
c_{17}	−52005112829	c_{36}	−54597476148812	c_{55}	−13634178346005		
c_{18}	36455433720	c_{37}	−55262827412672	c_{56}	−281488426825		
c_{19}	185715689189	c_{38}	33224040628261	c_{57}	2710991958599		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 73, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $73+4$ terms removed returns order 73 as well, so $\deg q_{\min} = 73 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $73+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 73 that this sequence satisfies — and that family turns out to have exactly one member.

References

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